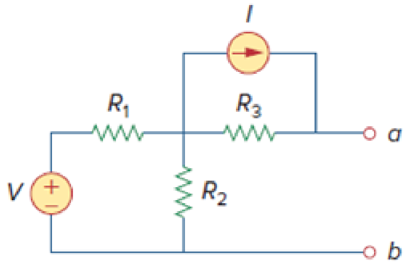
Chapter 4, Problem 49P

Problem

Find the Norton equivalent looking into terminals *a*-*b* of the circuit in Fig. 4.102. Let $V = 40$ V, $I = 3$ A, $R1 = 10\ \Omega$, $R2 = 40\ \Omega$, and $R3 = 20\ \Omega$.

Figure. 4.102:



Step-by-step solution

Step 1 of 5

To find R_{Th}, Consider the circuit in Fig-(a).

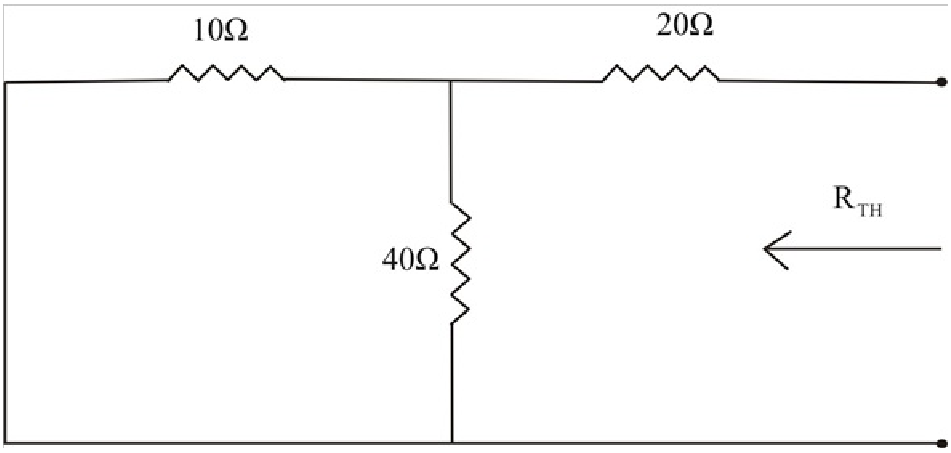


Fig-(a)

Step 2 of 5

$$R_N = 20 + (10 \| 40)$$

$$R_N = 20 + \frac{10 \times 40}{10 + 40}$$

$$R_N = 20 + \frac{400}{50}$$

$$R_N = 20 + 8$$

$$R_N = 28\Omega$$

Step 3 of 5

To find I_N, consider the Fig-(b)

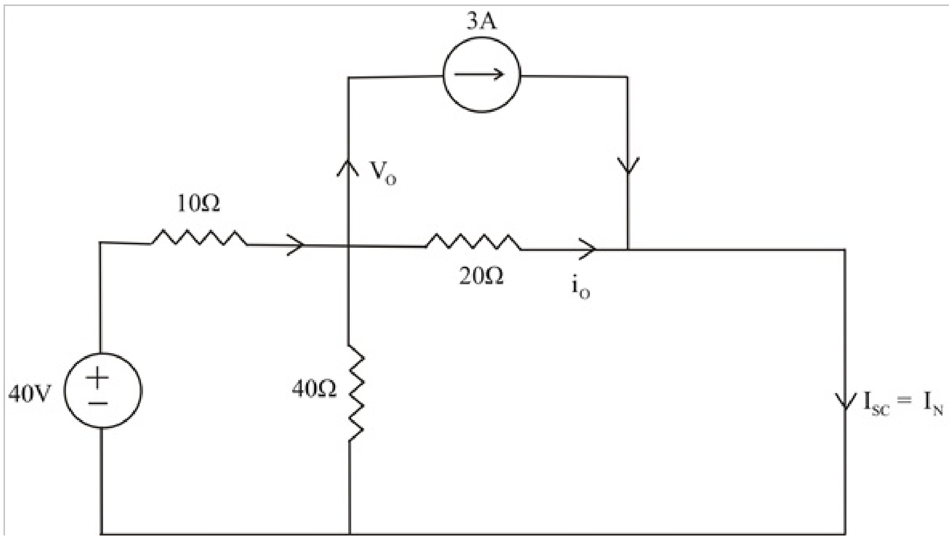


Fig-(b)

Step 4 of 5

At the node,

$$\frac{(40-V_0)}{10}=3+\left(\frac{V_0}{40}\right)+\left(\frac{V_0}{20}\right)$$

$$160-4V_0=120+V_0+2V_0$$

$$40=7V_0$$

$$V_0=\frac{40}{7}\text{V}$$

Therefore,

$$i_0=\frac{V_0}{20}$$

$$i_0=\frac{\frac{40}{7}}{20}$$

$$i_0=\frac{2}{7}\text{A}$$

But,

$$I_N=I_{SC}=i_0\ +\ 3$$

$$I_N=\frac{2}{7}+3$$

$$I_N=\frac{23}{7}$$

$$I_N=3.286\text{A}$$

Step 5 of 5

Therefore, the Norton equivalent circuit as shown in below;